Reinforcement Learning for SAM Prompts

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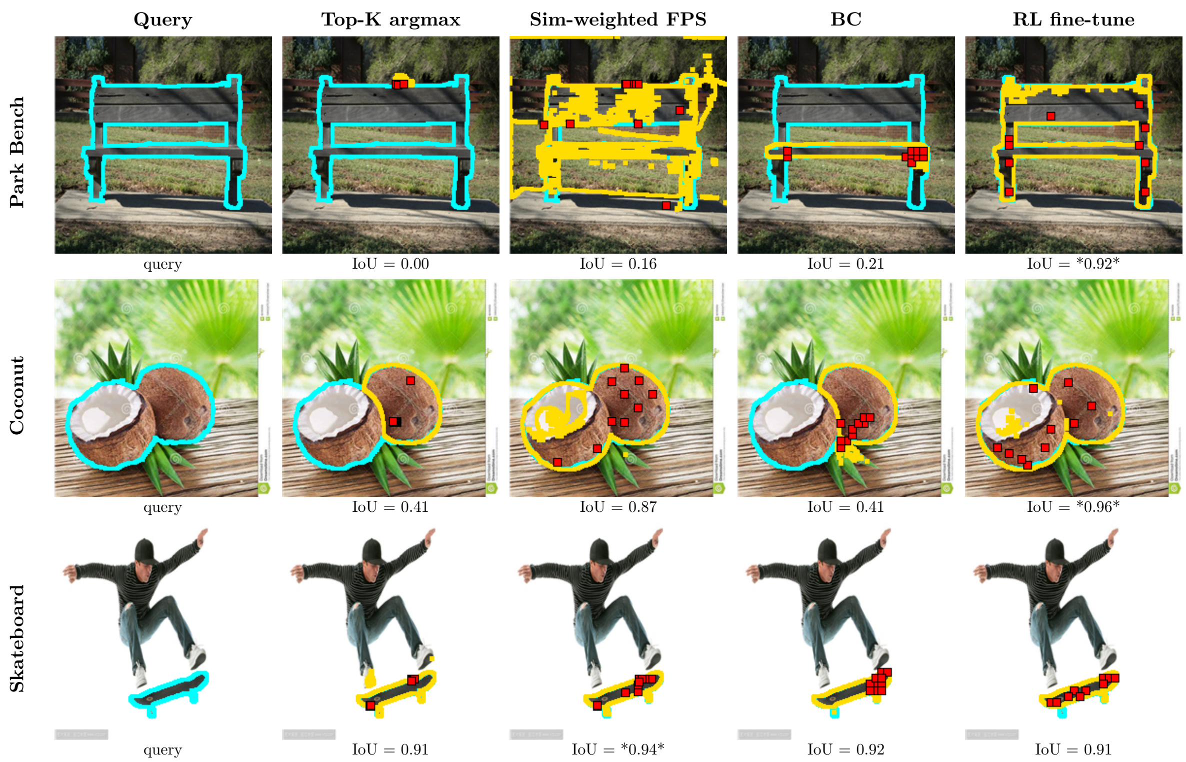

Fig. 1. Representative test-set masks with $N = 10$ on three FSS-1000 queries (rows) across four classes (columns). Yellow outlines mark the predicted SAM mask, cyan outlines mark the ground truth mask, and red dots mark the selected point-prompt positions (x, y); per-image IoU is printed below each panel. The RL fine-tune is the only method that recovers a complete mask on every query, while Top-K argmax fails outright on the park bench class, similarity-weighted FPS fragments under domain difficulty, and BC collapses on park bench and coconut tasks when longer horizons are required to reach a satisfactory position.

***Abstract*—Segmentation is a particularly time-consuming task. We believe that unsupervised point-prompt-based segmentation models can help speed up segmentation for new domains. The unsupervised models are class agnostic which means that human-supervised point-prompts are still required to get masks. Thus, we have evaluated point selection strategies for single-shot semantic segmentation under a shared frozen DINOv2 plus Segment Anything Model (SAM) pipeline on FSS-1000, comparing eleven heuristics, a behavioral-cloning policy, and a Proximal Policy Optimization (PPO) fine-tune. Under a unified action space, the strongest heuristic, similarity-weighted Farthest Point Sampling at $\alpha = 0.75$, achives a 0.768 mean IoU (mIoU) using 5 points ($N = 5$); behavioral cloning of a greedy oracle reaches 0.672 mIoU; and two reinforcement-learning (RL) fine-tunes warm-started from the cloned policy reach 0.864 mIoU at $N = 5$ and 0.866 mIoU at $N = 10$, with improvements of 0.193 and 0.281 over the behavioral cloning approach. On the canonical FSS-1000 one-shot subset, the same policies reach 0.876 and 0.875, surpassing PerSAM and SegGPT and approaching Matcher, which relies on the same DINOv2 plus SAM backbone.**

I. Introduction

Single-shot models learn to perform a task from a single labeled example. Single-shot semantic segmentation reframes this as a support–query problem. Given one annotated support image of a class, the model must segment every instance of that class in a new query image. The setting is attractive because it avoids the expensive per-class supervision that conventional segmentation networks require, and it aligns with how practitioners often operate in the wild, where a small number of reference annotations must generalize to unseen scenes. Recent progress in self-supervised representation learning [1] and promptable segmentation [2] has made this regime increasingly within reach for the average practitioner. In particular, the combination of a frozen vision transformer for feature extraction and the Segment Anything Model (SAM) for mask decoding has emerged as a dominant template, because it decouples class-agnostic shape priors

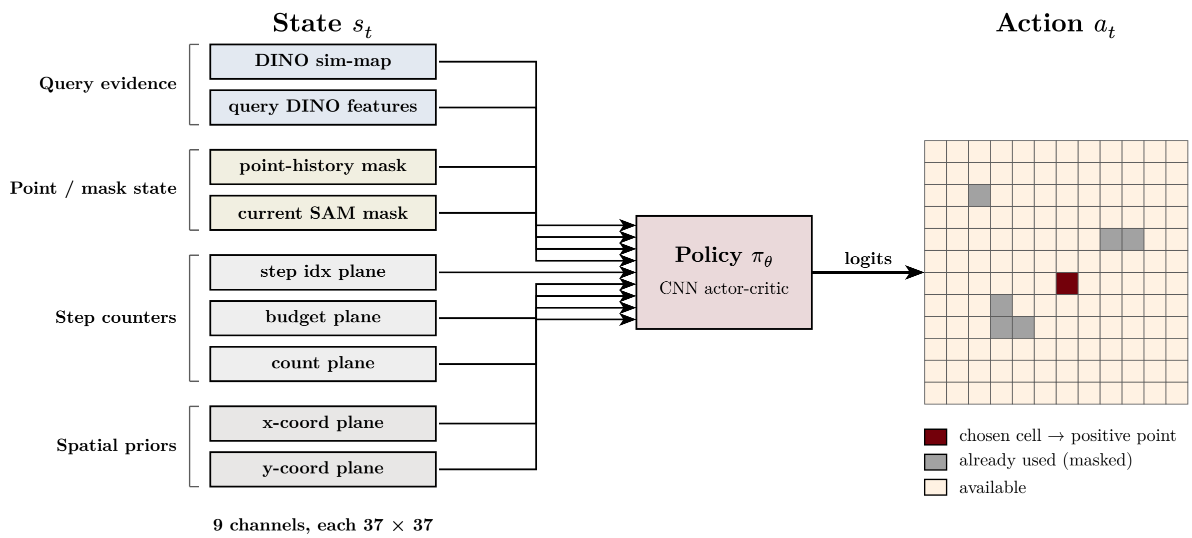

Fig. 2. Policy state and action representation. The state s_t is a nine-channel tensor at 37×37 resolution (DINOv2 similarity map, projected query features, point-prompt-history mask, current SAM mask, three step-counter planes, two coordinate planes), and the policy emits a logit over the same 37×37 grid with previous-point pixels masked out before sampling.

from semantic correspondence and requires no fine-tuning on the target domain.

The standard template for single-shot SAM-based segmentation involves three stages: 1) feature extraction via DINOv2, 2) similarity map computation, and 3) point prompt selection for SAM. DINOv2 embeds both support and query images into dense feature grids. The annotated support region is averaged to form a reference class embedding, which is compared to each query pixel via cosine similarity. Finally, N point prompts are extracted from the similarity map and passed to SAM, which produces the final mask.

However, this reduction is the most contested stage of the pipeline. The prior two stages have well-established closed-form definitions, but point selection from a dense similarity map has been approached in fundamentally different ways across the recent literature. The naive choice of selecting the N pixels with the highest similarity scores tends to concentrate points on a single high-confidence region of the object. SAM then receives redundant prompts that provide little additional disambiguating information about object extent, boundaries, or the presence of multiple instances, and the marginal value of additional points collapses.

In response, much of the recent SAM-prompting literature converges on the intuition that spatial spread, jointly with similarity magnitude, governs mask quality, and clustering-based heuristics including NMS, FPS, and k-means-style selectors have been proposed in service of that intuition [3], [4], [5]. Yet the spread framing is itself rarely defended empirically; a growing body of work argues that per-object considerations matter more than uniform spatial coverage [6], [7], [8], [9], and several methods bypass discrete point selection entirely in favor of dense correspondence or learned embeddings [10], [11]. These strategies are typically introduced as components of larger systems and compared only incidentally, leaving the relative merits of the underlying selection rules poorly characterized.

We address this gap with a benchmark of eleven non-trained point selection strategies, a behavioral-cloning (BC) policy, and a reinforcement-learning (RL) fine-tune, all evaluated on FSS-1000 [12] over $N \in \{1, 2, 3, 5, 7, 10\}$. All methods share an identical front-end and back-end, namely a DINOv2 ViT-L/14 encoder and a SAM ViT-H decoder, so that the only source of variation is from the N prompt locations that are drawn from the shared similarity map. To remove a residual source of confound, we additionally quantize every selector's output to a shared 37×37 grid before passing it to SAM, so that heuristic, imitation, and reinforcement-learning policies all operate in the same action space. Across this benchmark, the RL fine-tune outperforms the strongest spread-based heuristic by and breaks the plateau-and-decline pattern that every non-trained selector exhibits.

II. Related Work

Classical few-shot segmentation aggregates dense correspondences between a support image-mask pair and a query image, with a sequence of architectures progressively refining the aggregation strategy: prior-guided enrichment [13], hypercorrelation [14], base/meta separation [15], 4D cost aggregation [16], and dense cross-attention [17], [18]. These methods reach strong mIoU on FSS-1000 but require training on the few-shot task itself and assume the query mask can be regressed densely.

SAM [2] and SAM2 [19] made one-shot segmentation tractable from a small number of point or box prompts without per-class fine-tuning. Subsequent work pairs SAM with feature-matching for training-free few-shot segmentation [3], [20], conditions a single model on an in-context example [10], or adapts SAM2 with a trained adapter [21]. These

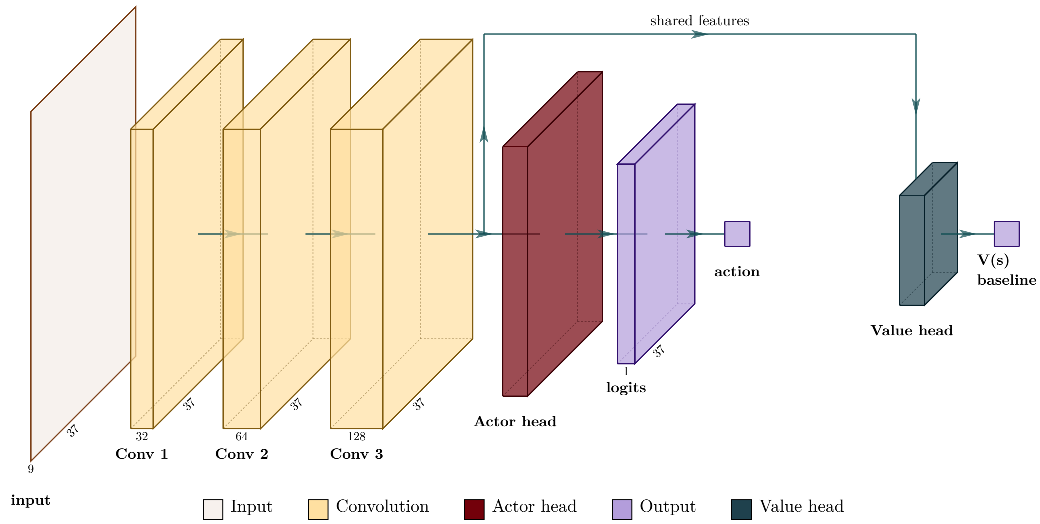

Fig. 3. Policy network architecture. Three convolutional blocks form a shared backbone over the nine-channel input. The actor head is a 1×1 convolution producing 37×37 logits, masked by previous-point-prompt pixels before sampling. The value head is a global-average pool followed by an MLP scalar baseline used by PPO's advantage estimator.

methods vary widely in how much information they pass to the segmentation backbone, leaving the prompt-selection rule itself underexplored.

A separate line of work trains networks to consume simulated point-clicks for interactive segmentation [22], [23], [24]. These methods operate in a fundamentally different setting from ours because the user (or simulator) provides the click sequence; the network is trained to consume points rather than produce them. To our knowledge, the present work is the first systematic comparison of heuristic, imitation, and reward-driven point selection on a unified action space against a frozen SAM decoder.

III. Method

A. Pipeline and notation

We benchmark thirteen point selection strategies under a shared single-shot semantic segmentation pipeline. Each episode consists of a support image with a support binary mask and a query image whose ground-truth mask is held out for evaluation. A frozen DINOv2 ViT-L/14 encoder maps both images to dense patch features at 224×224 resolution, and patch features are bilinearly upsampled to the input grid so that every pixel carries a feature vector. The support prototype $p \in \mathbb{R}^d$ is obtained by masked average pooling of the support feature map under the support mask, and the query similarity map is the cosine similarity between p and each query feature. We denote this map by $S \in [0,1]^{H \times W}$, where $H = W = 224$. The map is precomputed once per episode and cached, so every selection strategy receives the same input and any differences in downstream segmentation are attributed to point selection.

Given a fixed budget of point-prompts (N), a selection strategy is a deterministic or stochastic function that returns a point set $\mathcal{P}_N = \{(x_i, y_i)\}_{i=1}^{N}$ with $(x_i, y_i) \in \{1, \dots, W\} \times \{1, \dots, H\}$. The point set is passed to a frozen SAM ViT-H decoder, where every point is treated as a positive label and no negative labels or box prompts are supplied. The decoder returns a single binary mask that is compared against the ground truth mask. To make heuristic, imitation, and reinforcement-learning selectors directly comparable at the prompt level, every chosen pixel coordinate is quantized to the nearest center of a shared 37×37 grid before being passed to SAM. The grid resolution arises naturally from the BC oracle's candidate set, defined on this grid for tractable training, and we apply the same quantization to every selection rule so that all methods operate in a single action space.

B. Heuristic selectors

We evaluate eleven non-trained selection rules. *Top-K argmax* picks the N pixels with the highest values of the similarity map (S). *Top-K with non-maximum suppression (Top-K NMS)* iteratively appends the current $\arg\max$ of S to point set ($\mathcal{P}_N$) and then sets $S(x', y') = -\infty$ for every pixel within Euclidean radius r of the chosen point; we use $r = 14$. *Farthest Point Sampling (FPS)* seeds with the global $\arg\max$ of S and iteratively appends the pixel that maximizes the minimum Euclidean distance to all previously chosen points. *Similarity-weighted FPS* scores each candidate as $\alpha \cdot S(x,y) + (1-\alpha) \cdot \tilde{d}(x,y)$ with $\alpha = 0.75$, where $\tilde{d}$ is the normalized minimum distance to $\mathcal{P}$. *Top-K to k-means* takes the $K' = 256$ highest-similarity pixels, clusters them into N groups via k-means, and returns the highest-similarity pixel per cluster. *Local-maxima peaks* detects local maxima of S subject to a minimum-separation constraint of six pixels and returns the top N. *Connected-component centers* thresholds S at $\tau = 0.75$, extracts 8-connected components, and selects the $\arg\max$ of S within each component. *Mean-shift modes* runs mean-shift on similarity-weighted pixel coordinates with bandwidth six and returns the top N modes. *Quantile-spaced*

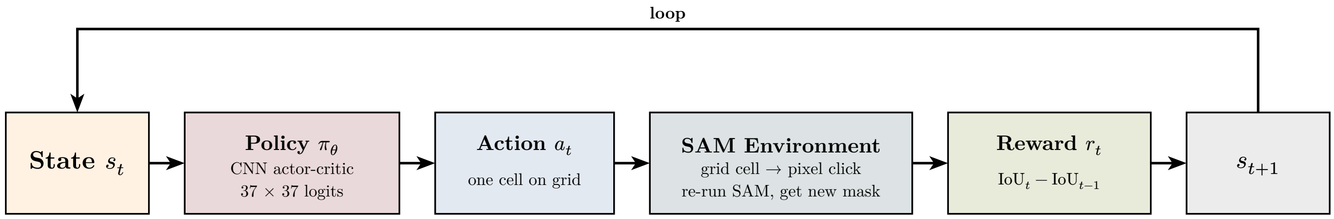

Fig. 4. Reinforcement-learning interaction loop. At each step the policy maps the multi-channel state to a 37×37 logit, samples a discrete point cell, passes it to SAM, and receives a reward equal to the change in IoU between the new and previous mask; the episode terminates after N points.

sorts pixels by similarity in descending order and picks at evenly spaced quantiles. *Similarity-weighted random* samples N points without replacement from the categorical distribution with probabilities $p(x, y) \propto \exp(S(x, y)/T)$ at $T = 0.05$. *Determinantal Point Process (DPP)* samples $\mathcal{P}_N$ from a k-DPP with quality scores $q_i = S(x_i, y_i)$ and a Gaussian diversity kernel of length-scale $\ell = 18$.

C. Behavioral-cloning policy

In addition to the eleven heuristics, we evaluate a learned policy trained by behavioral cloning (BC) on a greedy oracle. For each training episode, we discretize the 224×224 query grid to a 37×37 candidate grid and, for each candidate position (x_c, y_c), query SAM under the prompt set $\mathcal{P}_t \cup \{(x_c, y_c)\}$ and record the resulting IoU against the ground-truth mask. The oracle's t-th point is the candidate that maximizes this IoU. Because the oracle's first k points are identical for any $N \geq k$ by construction, a single $N = 10$ trajectory yields oracle labels for every smaller budget, and we cache trajectories per episode to memory so that the imitator never re-invokes SAM during training. On the FSS-1000 validation split this oracle reaches a mIoU 0.948 at $N = 2$, 0.952 at $N = 3$, 0.956 at $N = 5$, and 0.955 at $N = 10$, confirming that the 37×37 grid resolution is not the bottleneck and that substantial headroom exists above the heuristics.

The loss is a Gaussian-kernel-density formulation of cross-entropy in which the oracle target is rendered as a Gaussian heatmap with $\sigma = 2$ grid cells, so that the gradient signal degrades gracefully near the correct cell rather than treating any non-exact prediction as equally wrong. We add an auxiliary IoU loss term that, on a sub-batch of training examples, decodes the policy's argmax point and computes IoU against the ground-truth mask, providing a small, IoU-aligned supervisory signal. To match the inference-time setting, where re-clicking an already-clicked cell is meaningless, we mask previously-clicked cells out of the policy's logit space at training time. We train one policy per budget $N \in \{2, 5, 10\}$ on the FSS-1000 train classes, select checkpoints by full-validation rollout IoU on the 450 validation episodes, and evaluate on the held-out test split using the same eval setup as the heuristic baselines.

D. Reinforcement-learning fine-tune

The behavioral-cloning policy provides a strong initialization, but its training signal is the oracle's point identity rather than SAM's mask quality. To align the optimization signal with the actual evaluation criterion, we additionally fine-tune the cloned policy with reinforcement learning using SAM's IoU as the per-step reward. The setting is naturally formulated as a finite-horizon Markov decision process (MDP), illustrated in Fig. 4. The state at step t is $s_t = (Q, S, C_t, M_{t-1})$, where Q denotes the projected DINOv2 query features, S is the cosine similarity map, C_t is the binary click-history mask after $t-1$ placements, and M_{t-1} is SAM's mask under the prior prompt set. The action a_t is a discrete pixel on the same 37×37 grid the BC oracle was defined on, which preserves warm-start consistency. The reward at step t is the marginal IoU gain $r_t = \mathrm{IoU}(M_t, M^*) - \mathrm{IoU}(M_{t-1}, M^*)$, where M^* is the ground-truth mask.

We use proximal policy optimization (PPO) [25] with the categorical policy expressed as a softmax over the 37×37 logit map produced by the BC network. A linear value head is appended to the same backbone, yielding the shared-backbone actor-critic architecture in Fig. 3. The policy is initialized from the BC best checkpoint, and a Kullback-Leibler regularization term $\beta \cdot \mathrm{KL}(\pi_\theta \parallel \pi_{\mathrm{BC}})$ is added to the loss to anchor the fine-tune to the imitation prior; we initialize $\beta = 0.1$ and decay it linearly to 0.01 over the first quarter of training. The full warm-start-then-fine-tune pipeline is shown in Fig. 5. PPO collects rollouts from 32 vectorized environments, computes generalized advantages [26], and updates the policy with a clipped surrogate objective regularized by the KL anchor to the frozen BC reference.

We use standard PPO hyperparameters: clip $\varepsilon = 0.2$, GAE $\lambda = 0.95$, $\gamma = 1.0$, four update epochs per rollout, and an effective rollout batch of 32 vectorized environments times episode length. Across one million environment steps, the KL anchor coefficient β decays linearly from 0.1 to 0.01, the entropy coefficient from 0.01 to 0.001, and the learning rate from 3×10^{-4} to zero. Early steps keep the policy (π_θ) close to the BC reference; late steps relax the constraints and let the agent optimize the change in IoU directly. SAM ViT-B is used during PPO training to reduce per-call cost; final test evaluation uses the same SAM ViT-H decoder used by every other method, keeping the comparison unbiased.

We report two RL fine-tune runs, both warm-started from the BC policy at the matching budget and trained for one million environment steps. The $N = 5$ run targets the budget at which heuristic methods peak; the $N = 10$ run targets the longer horizon at which heuristic and imitation policies exhibit their characteristic late-decline failure mode.

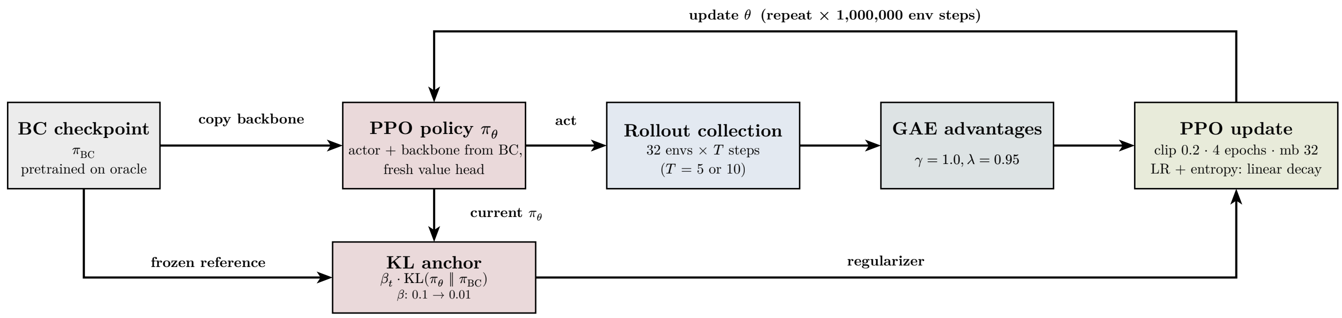

Fig. 5. BC warm-start to PPO fine-tuning. The actor weights are copied from the BC checkpoint and the value head is initialized fresh. PPO collects rollouts from 32 vectorized environments, computes generalized advantages, and updates the policy with a clipped surrogate objective regularized by a KL anchor to the frozen BC reference.

TABLE I
HEURISTIC VS. TRAINED VS. RL METHODS ON FSS-1000 TEST (450 EPISODES PER CELL).

Method	$N = 1$	$N = 2$	$N = 5$	$N = 7$	$N = 10$
Greedy Oracle	0.920	0.948	0.956	0.956	0.955
PPO RL	–	–	**0.864**	–	**0.866**
BC	–	0.634	0.672	–	0.586
FPS ($\alpha = 0.75$)	0.585	0.699	0.768	0.756	0.731
Top-K argmax	0.585	0.579	0.502	0.445	0.379

E. *Evaluation protocol*

We evaluate on FSS-1000 under a class-disjoint split in which the validation and test class pools are entirely disjoint, and we sample 450 episodes per split. Each episode is a tuple consisting of a support image, its support binary mask, a query image, and a query ground-truth mask, with the support mask and query image drawn from the same class. Selection strategies are evaluated at point budgets of $N = \{1, 2, 3, 5, 7, 10\}$. We perform grid-search on the validaiton set to tune each hueristic with tunable hyperparameters. Once the heuristic is tuned we freeze the weights and run inference for the test split. For stochastic methods we average over five seeds; deterministic methods run once with seed = 0.

To position our pipeline against external few-shot segmentation methods, we additionally evaluate on a 66-class subset of the canonical FSS-1000 one-shot benchmark [12] disjoint from our training set, yielding 660 test episodes under the canonical 10-queries-per-class protocol. External methods are re-evaluated on the same 660-episode subset using their respective public eval scripts, with their mIoU and foreground-background IoU computed on identical inputs to ours. As a sanity check, we re-ran Matcher on the full canonical 240-class test set and reproduced its originally reported mIoU of 0.872. We report intersection-over-union (IoU) as the primary metric.

IV. EXPERIMENTS

A. *FSS-1000 in-domain benchmark*

Table I and Fig. 6 report test-set mean IoU for all heuristics, the BC policy, and the RL fine-tune, evaluated at $N \in \{1, 2, 3, 5, 7, 10\}$ on 450 episodes per cell under the unified action space. Table III in the appendix lists every heuristic at every budget for completeness. The picture separates into clean tiers. The strongest heuristic, similarity-weighted FPS at $\alpha = 0.75$, reaches 0.768 mIoU at $N = 5$ and stays above 0.73 mIoU across $N \in \{3, 5, 7, 10\}$. Top-K NMS, Top-K to k-means, and local-maxima peaks track within 0.05 IoU of similarity-weighted FPS at every budget. A second tier composed of DPP, connected-component centers, and similarity-weighted random sits in the 0.65 to 0.69 range. Mean-shift modes and Top-K argmax form a weaker third tier, with Top-K argmax notably the only method whose IoU degrades monotonically and steeply with N, dropping 20.6 IoU points from $N = 1$ to $N = 10$ as redundant nearby points confuse SAM's prompt encoder. Pure FPS and quantile-spaced are catastrophic failures, falling below 0.20 IoU past $N = 1$ because they place points in low-similarity regions that lie off-object. Nine of the eleven heuristics report identical IoU at $N = 1$ (0.585), validating the implementation: with a single output point, all of these rules reduce to the argmax of the cosine-similarity map snapped to the nearest grid cell.

B. *Imitation policies*

The BC policy reaches 0.634 mIoU at $N = 2$, 0.672 mIoU at $N = 5$, and 0.586 mIoU at $N = 10$, narrowly underperforming the strongest heuristic at $N = 5$ by 0.096 IoU. The cluster-spread metric on BC's test rollouts confirms that train-time masking behaves as intended at small budgets but loses its effect with large budgets: the mean pairwise point distance is 47.9 pixels at $N = 2$ but contracts to 34.1 pixels at $N = 10$, indicating that the policy still drifts back into spatially correlated regions across multiple steps even when immediate re-clicking is forbidden.

C. *Per-step trajectory analysis*

Fig. 7 plots the per-step rollout IoU of BC and the PPO fine-tune at $N = 5$. The two curves diverge immediately. By step

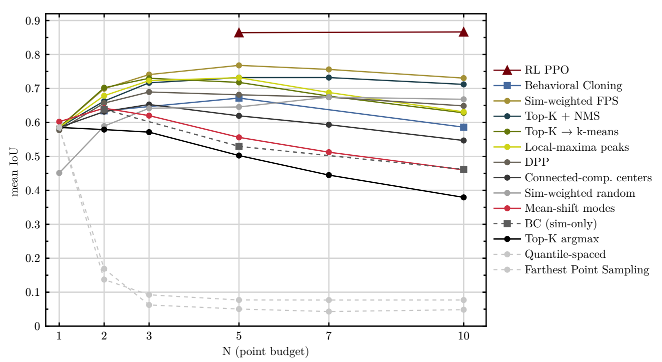

Fig. 6. Mean IoU on the FSS-1000 test set as a function of point budget N, under the unified 37×37 action space. The RL fine-tune (Garnet triangles) sits well above every other method at both trained budgets. The four Tier-1 heuristics cluster within 0.05 IoU of one another and peak at $N \in \{3, 5\}$. Top-K argmax decays monotonically; pure FPS and quantile-spaced collapse past $N = 1$.

2, RL reaches 0.786 mIoU while BC reaches 0.599 mIoU, and the RL trajectory is already above the strongest non-trained baseline shown as the dashed reference line. By step 3, RL is at 0.846 mIoU while BC is at 0.636 mIoU. The RL curve continues to rise through clicks 4 and 5 to its final value of 0.864, while the BC curve climbs only to 0.672 mIoU. The gap between the two trajectories at every step is at least 0.187 IoU.

The longer-horizon picture in Fig. 8 is sharper still. The RL fine-tune at $N = 10$ rises from 0.554 mIoU at step 1 to 0.777 mIoU at step 2, 0.831 mIoU at step 3, 0.852 at step 4, 0.863 mIoU at step 5, and reaches 0.868 mIoU at step 6. From step 6 onward, the trajectory sits on a flat plateau between 0.866 mIoU and 0.869 mIoU, with no sign of late decline. The BC curve on the same axes peaks at step 3 (0.645 mIoU) and degrades through every subsequent step to 0.586 mIoU at step 10. The plateau-and-decline shape is shared by every heuristic and by BC, suggesting that imitation alone cannot break a SAM-imposed ceiling. The RL trajectories' rise-then-plateau, sustained across both the short-horizon and long-horizon budgets, indicates that reward-driven training reshapes the per-step curve.

D. Reinforcement-learning fine-tune

The best PPO checkpoint at $N = 5$ occurs at update 5{, }250 of 6{, }250 with validation rollout IoU 0.903; the best checkpoint at $N = 10$ occurs at update 3{, }000 of 3{, }125 with validation rollout IoU 0.895. The mean pairwise point distance on test rollouts is 69.5 pixels at $N = 5$ and 69.3 pixels at $N = 10$, essentially identical across the two budgets, indicating that reward-driven training maintains the same wide point distribution as the budget grows rather than contracting toward a tight cluster as BC does (34.1 pixels at $N = 10$).

E. Comparison to external few-shot segmentation methods

Table II shows the results of our pipeline against four external one-shot segmentation methods on the 66-class canonical-disjoint subset. The PPO fine-tune at $N = 5$ reaches 0.876 mIoU on the canonical-disjoint subset, slightly higher than its performance on our in-domain test split (0.864). The PPO fine-tune at $N = 10$ reaches 0.875 mIoU. Among the external methods, SANSA leads at 0.923 mIoU and Matcher follows at 0.892 mIoU. Our RL fine-tune surpasses SegGPT (0.867 mIoU) and PerSAM (0.741 mIoU) but trails the two strongest external methods. The most informative single comparison is against Matcher, since the two methods share an identical DINOv2-L plus SAM-H backbone and differ only in how the SAM decoder is prompted: Matcher uses a training-free dense-correspondence prompting strategy that goes beyond the strict positive-point vocabulary, while our RL policy returns at most ten positive points. The 0.016 mIoU gap between the two indicates that, on the same backbone and the same canonical test, a learned point selector with a strict $N \leq 10$ point-prompt budget closely approaches the quality of a more expressive prompting strategy. The 0.047 mIoU gap to SANSA is best read as the cost of restricting the action space to grid-quantized positive points rather than as a deficiency of the RL training procedure itself, since SANSA replaces the entire encoder backbone with SAM2 and adds a trained adapter network.

TABLE II
COMPARISON AGAINST EXTERNAL FEW-SHOT METHODS ON THE CANONICAL FSS-1000 ONE-SHOT 66-CLASS DISJOINT SUBSET (660 EPISODES).

Method	mean IoU
SANSA (SAM2-Hiera-L + adapter) [21]	0.923
Matcher (DINOv2-L + SAM-H, training-free) [3]	0.892
SegGPT [10]	0.867
PerSAM [20]	0.741
RL fine-tune (PPO, $N = 5$)	0.876
RL fine-tune (PPO, $N = 10$)	0.875
Sim-weighted FPS ($\alpha = 0.75$, $N = 5$)	0.762
BC ($N = 5$)	0.641
Top-K argmax ($N = 5$)	0.474

The relative ordering among our methods is preserved on the canonical-disjoint subset. RL clears every heuristic by a wide margin. BC underperforms its in-domain numbers at the longer budgets (-0.031 at $N = 5$, -0.030 at $N = 10$) while RL slightly outperforms its in-domain numbers ($+0.012$ at $N = 5$, $+0.009$ at $N = 10$), suggesting that reward-driven policies are more robust to class-distribution shift than imitation policies trained from oracle points.

F. Ablations

We report two ablations using existing data, each isolating one design choice.

a) *Effect of warm-start initialization:*

The PPO fine-tune is warm-started from the BC best checkpoint and Kullback-Leibler (KL)-anchored to that same reference policy. Without the warm start, the policy network and the value head would both be initialized from scratch, and the KL anchor would reduce to a regularizer toward a degenerate prior. The relevant comparison in our data is between

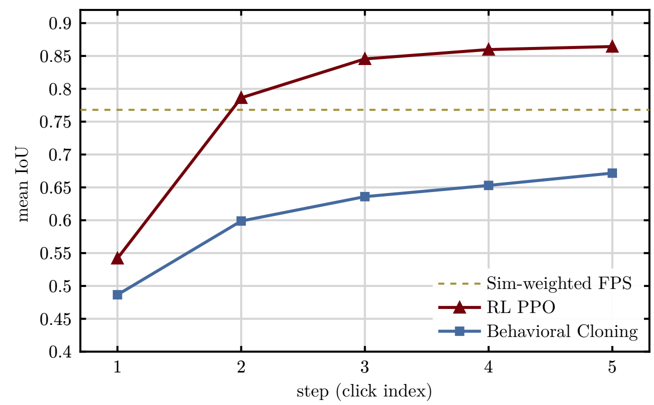

Fig. 7. Per-step rollout IoU at $N = 5$ on FSS-1000 test (450 episodes), comparing BC (Atlantic squares) against the PPO RL fine-tune warm-started from BC (Garnet triangles). The RL trajectory rises monotonically across all five points and clears the strongest non-trained baseline (dashed reference at 0.768) by step 2, while BC stays below the baseline at every step.

the BC starting point (0.672 mIoU at $N = 5$) and the PPO converged checkpoint (0.864 mIoU at $N = 5$), demonstrating that PPO recovers an additional 0.193 IoU starting from a non-degenerate prior. The KL anchor at $\beta = 0.1$ decaying to 0.01 keeps the policy close to the BC reference early in training, when the value-function estimate is still noisy and unconstrained gradient steps would be most likely to destroy the warm-start signal.

b) *Effect of action-space discretization:*

The 37×37 grid discretization could be considered as an artificial constraint. Thus, we compare the heuristic numbers under the grid-snapped protocol against the same heuristics evaluated at full 224×224 pixel resolution on our in-domain split. The strongest heuristic, similarity-weighted FPS at $\alpha = 0.75$, scores 0.759 mIoU at $N = 5$ at full resolution and 0.768 at $N = 5$ under grid quantization, a shift of $+0.009$ IoU. The grid-snapping step slightly improves the heuristics on average because it enforces a minimum spacing between selected pixels, partially solving the redundant-points problem on its own. The Top-K argmax decay shape is preserved: at $N \in \{1, 2, 5\}$, the full-resolution IoU sequence is $(0.594, 0.586, 0.499)$ versus $(0.585, 0.579, 0.502)$ under grid quantization. Thus , the win for RL is not an artifact of the discretization. The grid step alone fails to close the gap to 0.864 mIoU that the RL policy reaches under the same protocol.

G. Cross-domain smoke test

We additionally ran a small cross-domain smoke test on Kvasir-SEG, a polyp segmentation dataset from medical imaging. Ten random support-query image pairs were sampled, with no class-disjoint structure (the dataset has only the polyp class). The same DINOv2-L sim-map and SAM ViT-H decoder were used; the heuristic, BC, and RL policies were applied without retraining. The strongest heuristic, similarity-weighted FPS at $N = 5$, reached only 0.149 mIoU on the ten pairs. The BC policy reached 0.357 mIoU and the RL fine-tune reached 0.339 mIoU. All three values represent substantial degradation relative to FSS-1000, consistent with significant out-of-distribution shift from natural-image few-shot data to medical imagery. The heuristic crashed the hardest because it relies entirely on the DINOv2 cosine similarity map, which itself degrades under domain shift; the learned policies (BC and RL), which take richer DINOv2 query features as additional input, degrade more gracefully but show no clear RL-over-BC advantage in this OOD setting.

H. Representative examples

The visual contrast confirms the results of Table I: the RL fine-tune is consistently competitive, BC has individual-episode failure modes at long horizons, and heuristics depend heavily on the scene.

I. Inference cost

Wall-clock cost is not the limiting factor for any of the strategies considered. Most heuristic methods run in under thirty milliseconds per call on a single GPU at the resolutions used here. The slowest heuristic case is the DPP with $\ell = 18$ at $N = 10$, which reaches roughly forty-five milliseconds because of the eigen decomposition of its kernel. The fastest are Top-K argmax and quantile-spaced selection at approximately seven milliseconds. The BC and RL policies' per-step inference cost is dominated by the SAM decoder call itself; the policy network's contribution is a small additional CNN forward pass that runs in single-digit milliseconds.

V. Discussion

A. Imitation hits a ceiling, reward breaks it

The clearest empirical finding of this work is the contrast between the per-step trajectory shapes of imitation and reward-driven policies. The behavior of Top-K argmax is the cleanest diagnostic: its mIoU falls monotonically from 0.585 mIoU at $N = 1$ to 0.379 mIoU at $N = 10$, a drop of 20.6 points across the budget range. Confidence-only selection draws its top points from a tight neighborhood around the global similarity maximum, and adding more such points does not enrich the prompt set. SAM's prompt encoder is designed to consume each point as an additional disambiguating constraint on the latent mask, but redundant points at essentially the same spatial location carry no new disambiguation. The four Tier-1 heuristics, which enforce spatial separation in different ways, all converge to within 0.05 IoU at their respective peaks, suggesting that what SAM wants from its prompt set is decorrelation rather than any specific separation rule.

The behavioral-cloning policy provides a more interesting case. The greedy oracle that BC imitates is itself SAM-aware, in the sense that each oracle point is selected by maximizing IoU under SAM. Yet a policy trained to clone the oracle's click identities does not match the oracle's per-step trajectory. The cloned policy reproduces a pattern of plateau and decline rather than the oracle's continued ascent, peaking at point step 3 at $N = 10$ and degrading thereafter. Imitation alone, even of a strong oracle, appears to inherit the heuristic-style ceiling rather than break through it.

The plateau-and-decline that bounds the heuristic family and the imitation policy is broken cleanly by reward-driven training. The operative difference is the alignment of the optimization signal. Imitation forces the policy to commit to the specific cells the oracle chose at each step, which is brittle on the test distribution; reward-driven training lets the policy choose any cell whose click happens to improve SAM's mask, and over many gradient steps that flexibility lets the policy discover point sequences whose joint information content exceeds anything any single oracle trajectory exhibits.

B. Spatial spread, similarity floor, and action-space effects

The catastrophic failures of pure FPS and quantile-spaced selection sharpen this picture from the opposite direction. Both collapse below 0.20 mIoU past $N = 1$. After the first seed, pure FPS optimizes spatial distance with no regard for feature similarity, and on the small objects typical of FSS-1000 *far from chosen points* usually means *outside the object*. SAM treats every point as a positive label, so it grows the mask outward to encompass the resulting background points. Quantile-spaced selection has the same defect by construction. The general lesson is that any selector must enforce a similarity floor: every chosen point must lie in a region whose feature similarity is consistent with the support prototype, regardless of whether the rule is heuristic or learned.

The cluster-spread metric tells a complementary story for the learned policies. The mean pairwise point distance under RL is 69.5 pixels at $N = 5$ and 69.3 pixels at $N = 10$, essentially invariant across budgets. Under BC, the same metric contracts from 47.9 pixels at $N = 2$ to 34.1 pixels at $N = 10$, indicating that the imitation policy gradually re-introduces redundancy as the budget grows even when train-time masking explicitly penalizes immediate re-clicking. Reward-driven training discovers a budget-aware placement strategy; imitation discovers a fixed pattern that fails to use the longer horizon productively. The action-space ablation confirms this is not a discretization artifact: the heuristic numbers shift by less than 0.01 IoU when the same selectors are evaluated under the unified 37×37 grid versus full pixel resolution.

C. Limitations and future work

The benchmark was conducted on FSS-1000 alone, and the cross-domain smoke test on Kvasir-SEG suggests that all three method classes degrade substantially under medical-domain shift, with the heuristic family degrading hardest because of similarity-map degradation. We leave full medical-domain evaluation, including potentially domain-adapted features and proper episode protocols, to future work. All points in this study were treated as positive labels; neither negative-point prompts nor bounding-box prompts were explored, although both are important in SAM's prompt vocabulary. Extending the policy's action space to include negative points is a particularly natural next step because negative points are the standard way to correct over-segmentation, and the present action space cannot issue them. The class-disjoint split rules out within-class memorization but does not probe cross-domain generalization at scale; a multi-dataset evaluation including Pascal-5^i and COCO-20^i would strengthen the headline claim. Finally, we report a single PPO seed per budget; multi-seed

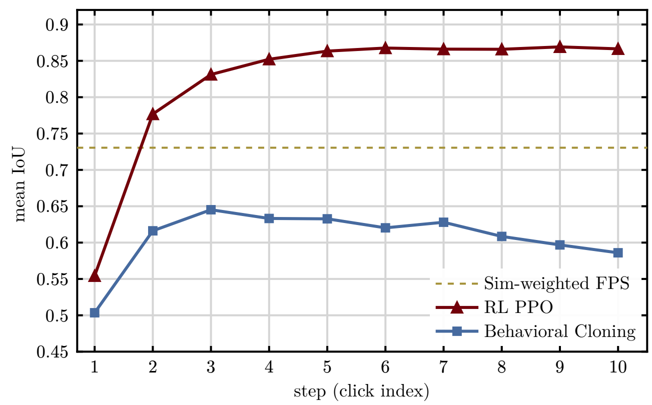

Fig. 8. Per-step rollout IoU at $N = 10$ on FSS-1000 test, comparing BC against the RL fine-tune at $N = 10$. BC peaks at point 3 (0.645 mIoU) and declines monotonically through step 10 (0.586 mIoU); RL rises sharply through step 6 and then sits on a flat plateau between 0.866 and 0.869 mIoU through step 10, sustaining its peak quality across the entire long-horizon budget.

stability and the dependence on warm-start quality remain to be characterized.

VI. Conclusion

We presented a controlled comparison of point selection strategies for single-shot semantic segmentation under a shared DINOv2 and SAM pipeline on FSS-1000, with all selectors operating in a unified 37×37 action space so that heuristic, imitation, and reinforcement-learning methods are directly comparable at the prompt level, and we additionally benchmarked the best of these methods against four external few-shot segmentation systems on the canonical FSS-1000 one-shot test protocol restricted to a 66-class subset disjoint from our training set. The reinforcement-learning policies rise sharply through their first six points and then sit on a high plateau, breaking the plateau-and-decline shape that every other method exhibits. The empirical claim of the paper is that imitation alone cannot break the SAM-imposed ceiling that bounds the heuristic family, but that aligning the optimization signal with SAM's actual mask quality, via reinforcement learning warm-started from a behavioral-cloning prior.

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I. Appendix

TABLE III
Full heuristic sweep on FSS-1000 test (450 episodes per cell), under the unified action space. Frozen-best hyperparameters per method.

Method	$N=1$	$N=2$	$N=3$	$N=5$	$N=7$	$N=10$
Sim-weighted FPS ($\alpha = 0.75$)	0.585	0.699	0.741	0.768	0.756	0.731
Top-K + NMS ($r = 14$)	0.585	0.663	0.717	0.732	0.732	0.712
Top-K to k-means ($K' = 256$)	0.585	0.702	0.730	0.718	0.677	0.628
Local-maxima peaks (sep = 6)	0.585	0.679	0.723	0.732	0.688	0.631
DPP ($\ell = 18$)	0.577	0.657	0.690	0.681	0.674	0.649
Connected-component centers ($\tau = 0.75$)	0.585	0.632	0.653	0.619	0.593	0.547
Sim-weighted random ($T = 0.05$)	0.451	0.589	0.642	0.646	0.674	0.668
Mean-shift modes (bw = 6)	0.602	0.642	0.620	0.556	0.512	0.460
Top-K argmax	0.585	0.579	0.571	0.502	0.445	0.379
Quantile-spaced	0.585	0.137	0.092	0.077	0.077	0.077
Farthest Point Sampling	0.585	0.169	0.062	0.050	0.043	0.048